CURRICULUM VITAE

Chiara Paola Montagna

Personal data

address	c/o Istituto Nazionale di Geofisica e Vulcanologia sezione di Pisa via Cesare Battisti, 53 56125 Pisa Italy
phone	+39 0508311947
email	chiara.montagna@ingv.it
date of birth	26.08.1979
place of birth	Milano
citizenship	Italian

Research interests

Magma thermo-fluid dynamics in volcanic systems: physical and numerical modeling.

Volcano deformation: physical and numerical modeling, monitoring data analysis and interpretation.

Fluid dynamics of compressible and incompressible multi-phase, multi-component flows; development of specific hpc software to target magma dynamics.

Current working position

Researcher at Istituto Nazionale di Geofisica e Vulcanlogia (INGV, National Institute for Geophysics and Volcanology), sezione di Pisa, Italy.
Member of European Geosciences Union (EGU), American Geophysical Union (AGU) and International Association of Volcanology and Chemistry of the Earth's Interior (IAVCEI) and Associazione Italiana di Vulcanologia (AIV).

Work experience

10.2007 - 02.2008	Contract researcher at INGV, Pisa.
03.2008 - 12.2011	Post-doctoral researcher at INGV Pisa.
01.2012 - 02.2013	Post-doctoral research associate at Rice University Department of Earth Science, Houston, TX, U.S.A..

Education

1998	High School Diploma (scientific studies), 60/60, Liceo scientifico "F. Lussana", Bergamo.
24.03.2003	Degree in Physics, 110/110 cum laude, Università degli Studi di Pisa.
22.09.2003	Master Degree in Physics, 110/110 cum laude, Università degli Studi di Pisa. Thesis *"Drift-kinetic theory of magnetic reconnection in collisionless plasmas"*, under the supervision of Prof. Francesco Pegoraro.
15.10.2008	Ph.D. Degree in Applied Physics, at the Physics Department of the University of Pisa, with the thesis *"Stationary plasmas in Vlasov theory"*, under the supervision of Prof. Francesco Pegoraro.

Languages

Italian	Mother tongue.
English	Fluent.
German	Good.

Informatics

Operating systems	Linux, MacOS and Windows.
Applications	Good knowledge of the most common software (document editing, internet browsing, email clients).
Programming languages	FORTRAN77, FORTRAN90, C++, python, R, MATLAB and Mathematica.

Peer reviewed publications

M. Trolese, A. Tadini, L. Pieretti, D. Biagini, S. Cianetti, S. Colucci, M. Cerminara, C. D'Oriano, C.P. Montagna, M. D'Ambrosio, R. Pegna, G. Re, F. Sanseverino, C. Giunchi, C. Meletti, T. Esposti Ongaro (2025) *"Evaluating the effectiveness of quantitative descriptions of Earth Science phenomena during outreach activities"*, Geoscience Communication, 8, 319–337. doi:10.5194/gc-8-319-2025

C. D'Oriano, C.P. Montagna, S. Colucci,P. Del Carlo, F. Brogi, D. Morgavi, A. Musu, F. Arzilli, S. Costa, P. Landi (2025) *"Fe-rich filamentary textures reveal timescales of magmatic interaction before the onset of high-energy explosive events at basaltic volcanoes"*, Volcanica, 8(1), 159–174. doi: 10.30909/vol/wytv2139

F. Martinelli, T. Esposti Ongaro, C.P. Montagna, (2025) *“VO-CIV: A web service solution for sharing geo scientific software.*, Rapporti Tecnici INGV, 495: 1-28. doi:10.13127/rpt/495

L. Insolia, S. Guerrier, C.P. Montagna, M.-P. Victoria-Feser, L. Caricchi (2024) *“Estimation and uncertainty quantification of magma interaction times using statistical emulation”*, Volcanica 7(2), 525. doi:10.30909/vol.07.02.525539

C. Pelullo, S. Chakraborty, C.P. Montagna, I. Arienzo, R.J. Brown, M. D’Antonio, S. de Vita, C. D’Oriano, M. Nazzari, L. Pappalardo, P. Petrosino (2024) *“A multi-methodological approach to record dynamics and timescales of the plumbing system of Zaro (Ischia Island, Italy)”*, Contributions to Mineralogy and Petrology 179, 54. doi:10.1007/s00410-024-02138-9

L. Caricchi, C.P. Montagna, A. Aiuppa, J. Lages, G. Tamburello, P.Papale (2024) *“CO_2 flushing triggers paroxysmal eruptions at open conduit basaltic volcanoes.”*, Journal of Geophysical Research: Solid Earth, 129, e2023JB028486.https://doi.org/10.1029/2023JB02

S. Colucci, F. Brogi, C.P. Montagna, G. Sottili, P. Papale, *“Short-term magma-carbonate interaction: A modelling perspective ”*, Earth and Planetary Letters 628, 118592 (2024). doi: 10.1016/j.epsl.2024.118592

D. Carbone, F. Cannavò, C.P. Montagna, F. Greco, *“Gas buffering of magma chamber contraction during persistent explosive activity at Mt. Etna volcano”*, Communications Earth and Environment 4, 471 (2023). doi:10.1038/s43247-023-01149-x

A. Longo, D. Garg, P. Papale, C.P. Montagna, *“Dynamics of magma chamber replenishment under buoyancy and pressure forces.”*, Journal of Geophysical Research: Solid Earth, 128, e2022JB025316 (2023). doi:10.1029/2022JB025316

F. Brogi, S. Colucci, J. Matrone, C.P. Montagna, M. de’ Michieli Vitturi, P. Papale, *“MagmaFOAM-1.0: a modular framework for the simulation of magmatic systems”*, Geoscientific Model Development, 15, 3773–3796 (2022). doi:10.5194/gmd-15-3773-2022.

C.P. Montagna, P. Papale, A. Longo, *“Magma Chamber Dynamics at the Campi Flegrei Caldera, Italy.”* In: G. Orsi, L. Civetta, M. D’Antonio (eds). Campi Flegrei. Springer (2022). doi:10.1007/978-3-642-37060-1_7

M. Bagagli, C.P. Montagna, P. Papale, A. Longo, *“Signature of magmatic processes in strainmeter records at Campi*

Flegrei (Italy)”, Geophysical Research Letters 44(2), 718 (2017).

S. Colucci, P. Papale, C.P. Montagna, *“Non-Newtonian flow of bubbly magma in volcanic conduits”*, Journal of Geophysical Research: Solid Earth (2017). doi:10.1002/2016JB013383.

P. Papale, C. P. Montagna, A. Longo, *“Pressure evolution in shallow magma chambers upon buoyancy-driven replenishment”*, Geochemistry, Geophysics, Geosystems, doi:10.1002/2016gc006731 (2017).

C. P. Montagna, P. Papale, A. Longo, M. Bagagli, *“Magma chamber rejuvenation: insights from numerical models”*. In: J. Gottsmann, J. Neuberg, B. Scheu (eds). Volcanic Unrest - From Science to Society. Springer (2017).

D. Morgavi, I. Arienzo, C.P. Montagna, D. Perugini, *“Magma mixing: history and dynamics of an eruption trigger”*. In: J. Gottsmann, J. Neuberg, B. Scheu (eds). Volcanic Unrest - From Science to Society. Springer (2017).

M. Polacci, M. De' Michieli Vitturi, et al., *“From magma ascent to ash generation: Investigating volcanic conduit processes by integrating experiments, numerical modeling, and observations”*, Annals of Geophysics 60(6), S066, doi:10.4401/ag-7449 (2017).

C. P. Montagna, P. Papale, A. Longo, *“Timescales of mingling in shallow magmatic reservoirs”*. In: L. Caricchi, J.D. Blundy (eds), Chemical, Physical and Temporal Evolution of Magmatic Systems, Geological Society, London, Special Publications 422, 6 (2015).

C. P. Montagna, H. M. Gonnermann, *“Magma flow between summit and Pu‘u ‘Ō‘ō at Kīlauea Volcano, Hawai‘i*, Geochemistry, Geophysics, Geosystems 14, 2232 (2013).

A. Longo, P. Papale, M. Vassalli, G. Saccorotti, C. P. Montagna, A. Cassioli, S. Giudice, E. Boschi, *“Magma convection and mixing dynamics as a source of Ultra-Long-Period oscillations”*, Bulletin of Volcanology 74 (4), 873 (2012).

M. Vassalli, A. Longo, C. P. Montagna, G. O'Brien, C. Bean, L. Bisconti, P. Papale, G. Saccorotti, *“An integrated method to model volcanic processes and associated geophysical signals”*. In: C. J. Bean, A. K. Braiden, I. Lokmer, F. Martini, G. S. O'Brien (eds). The VOLUME project, VOLcanoes: Understanding subsurface mass moveMEnt. Dublin: The Volume Consortium (2009).

A. Longo, M. Barsanti, P. Papale, M. Vassalli, C. P. Montagna, L. Bisconti, G. Saccorotti, *"A numerical code for the simulation of magma-rocks dynamics"*, Communications to SIMAI Congress 3, 237 (2009).

C. Montagna, F. Pegoraro, *"Vlasov equilibria: varying the temperature or the density distributions"*, Communications in Nonlinear Science and Numerical Simulation, 13 (1), 147 (2008).

C. Montagna, F. Pegoraro, *"Vlasov-Maxwell plasma equilibria with temperature and density gradients: Weak inhomogeneity limit"*, Physics of Plasmas 14, 042103 (2007).

F. Ceccherini, C. Montagna, F. Pegoraro, G. Cicogna, *"Two-dimensional Harris-Liouville plasma kinetic equilibria"*, Physics of Plasmas 12, 52506 (2005).

Preprints

S. Colucci, F. Brogi, C.P. Montagna, G. Sottili, P. Papale, *"Short-term magma-carbonate interaction: A modelling perspective "*, EarthArXiv (2023). doi:10.31223/X56M40

L. Caricchi, C.P. Montagna, A. Aiuppa, J. Lages, G. Tamburello, P. Papale, *"CO_2 flushing triggers large eruptions at open conduit volcanoes: the case of Stromboli (Italy)"*, Research Square (2023). doi:10.21203/rs.3.rs-2488366/v1

L. Insolia, S. Guerrier, C.P. Montagna, M.-P. Victoria-Feser, L. Caricchi, *"Estimation and Uncertainty Quantification of Magma Interaction Times using Statistical Emulation"*, EarthArXiv (2022). doi:10.31223/X5D36F.

C.P. Montagna, P. Papale, *"Time scales of shallow magma chamber replenishment at Campi Flegrei caldera."*, EarthArXiv (2018). doi:10.31223/osf.io/kgt5p

Software

C.P. Montagna, `pyFlushing`: A python driver to simulate CO_2 flushing in a magma column through equilibrium steps. Source code.

F. Brogi, S. Colucci, C.P. Montagna, `MagmaFOAM`: a modular framework for the simulation of magmatic systems based on the open source computational fluid dynamics software OpenFOAM. Release.

S. Colucci, C.P. Montagna, `syntheticSignals`: a suite of MATLAB scripts to produce and analyze synthetic geophysical signals from complex underground sources. Releases, source code.

Datasets

C.P. Montagna, Pre-eruptive mingling simulations - Zaro eruption, Ischia doi:10.5281/zenodo.11085722

J. Ruz-Ginouves, S. Colucci, F. Brogi, C.P. Montagna, FLOW-FOCUS - Simulations of flow in a fissure doi:10.82554/sdl-338

C.P. Montagna, R. Bruni, D. Garg, P. Papale, DT-GEO gales scenarios dataset DT5101 - Ground deformation simulations for dike intrusions at Mount Etna volcano doi:10.82554/sdl-115

Funding

10.2007 - 02.2009 ERC - 6th Framework Programme, *"VOLUME. VOLcanoes: Understanding subsurface mass moveMEnt"*, scientific coordinator C. Bean (University College Dublin).

05.2008 - 07.2010 INGV-DPC 2007-2009, *"V1 - Unrest: Realization of an integrated method for the definition of the unrest phases at Campi Flegrei"*, scientific coordinators L. Civetta (Università di Napoli) e E. Del Pezzo (INGV Napoli - Osservatorio Vesuviano).

05.2008 - 07.2010 INGV-DPC 2007-2009, *"V4 - Flank"*, scientific coordinators V. Acocella (Università di Roma Tre) e G. Puglisi (INGV Catania).

15.03.2008 - 15.03.2011 MIUR-FIRB *"AIRPLANE - Piattaforma di ricerca multi-disciplinare su terremoti e vulcani"*, scientific coordinator A. Neri (INGV Pisa).

11.2011 - 11.2015 ERC - 7th Framework Programme, *"VUELCO. Volcanic unrest in Europe and Latin America: Phenomenology, eruption precursors, hazard forecast, and risk mitigation"*, scientific coordinator J. Gottsmann (University of Bristol).

01.2012 - 01.2016 ERC - 7th Framework Programme, *"NEMOH - Numerical, Experimental and stochastic Modelling of vOlcanic processes and Hazard: an Initial Training Network for the next generation of European volcanologists"*, Marie Curie Initial Training Network, scientific coordinator P. Papale (INGV Pisa).

2012 - 2015 INGV-DPC 2012-2015, *V2: Eruption precursors*, scientific coordinators G. Chiodini (INGV Naples) and V. Acocella (Università di Roma Tre).

06.2013 - 07.2016 ERC - 7th Framework Programme, *"MED-SUV - Mediterranean Supersite Volcanoes"*, Collaborative project, scientific coordinator G. Puglisi (INGV Catania); responsible for sub-task 4.3.1: Modelling of internal dynamics of magmatic sources and their effect on surface deformation and gravity field.

02.2017 - 02.2020 MIUR-PRIN (Italian Government) 'Advanced atomic interferometer for gravity and quantum physics experiments with applications to geophysics', co-PI, €56500.

2018 INGV FISR (internal call) 'Time scales of magma chamber evolution', PI, €9346.

2021 - 2023 INGV Pianeta Dinamico (internal call) *"CHOPIN - volCano-atmospHere-iOnosPhere connectIoN"*, co-PI, €25800.

01.09.2022 - 31.08.2025 EU - H2020, *"IMPROVE - Innovative Multi-disciplinary European Research training network on VolcanoEs"*, Marie Sklodowska-Curie European Training Network, scientific coordinator P. Papale, INGV Pisa.

01.09.2022 - 31.08.2025 EU - HorizonEurope *"DT-GEO - A Digital Twin for GEOphysical extremes"*, Infrastructural project, scientific coordinator R. Carbonell, CSIC Barceola, Spain.

01.10.2022 - 30.09.2026 EU - HorizonEurope *"Geo-INQUIRE - Geosphere INfrastructures for QUestions into Integrated REsearch"*, Infrastructural project, scientific coordinator F. Cotton, GFZ Potsdam, Germay.

01.09.2024 – 31.08.2028 Swiss National Science Founration - Sinergia *"VAMOS - Data-driven imaging of volcanic plumbing systems "*, co-PI, €121900.

01.11.2026 - 30.10.2030 HORIZON-MSCA-2025-SE-01 *"REALI-3SE - Bridging Igneous Petrology and Machine Learning through Intersectoral, Interdisciplinary, and Intercontinental Staff Exchanges "*, co-PI, €90180.

Seminars and invited presentations

13.01.2008 Department of Physics, University of York, U.K.. *"Stationary plasmas in Vlasov theory."*

17.05.2010 Department of Physics, University of Pisa, Italy. *"Physico-mathematical models of magma fluid dynamics in volcanic systems."*

15.07.2010 Department of Physics, Centre for Fusion, Space and Astrophysics, University of Warwick, U.K.. *"Stationary plasmas in Vlasov theory."*

22.10.2010	Department of Geophysics, University of Hamburg, Germany. *"Magma convection and mixing in complex reservoirs as source of geophysical signals."*
22.11.2011	Istituto Nazionale di Geofisica e Vulcanologia, sezione di Pisa. *"Simulazioni numeriche di mixing di magmi e segnali geofisici associati"*.
4.9.2012	USGS Hawaiian Volcano Observatory, *"Magma flow between summit and Puʻu ʻŌʻō at Kīlauea Volcano, Hawaiʻi"*.
11.12.2018	AGU Fall Meeting, C.P. Montagna, P. Papale, M. Bagagli, G. Saccorotti, *"Deciphering Volcanic Unrest through Forward-Modeled Monitoring Datasets"*.
14.12.2021	AGU Fall Meeting, C.P. Montagna, L. Caricchi, *"Messengers from Depth: Impact of Fluid-Magma Interaction on Gas Measurements and Eruptive Dynamics at Active Volcanoes"*.
25.6.2025	GeoSeminar DIAS Dublin Institute of Advanced Studies, School of Cosmic Physics - Geophysics, *"Modeling magma flow dynamics at different length and time scales"*.
15.12.2025	AGU Annual Meeting, C.P. Montagna, R. Bruni, E. De Paolo, M. Allegra, D. Garg, F. Cannavò, P. Papale, *"A Digital Twin for identifying unrest and dike propagation patterns at Mount Etna volcano, Italy"*. More than 50 presentations at international conferences.

Advising

2025-2028	Davide Bonaretti, PhD student at the Department of Physics, University of Pisa;
2022-2025	Owen McCluskey, PhD student at the Department of Earth Sciences, University of Pisa; EU Horizon Europe ITN fellow within the IMPROVE project.
2022-2025	Gabriel Girela Arjona, PhD student at the Department of Earth Sciences, University of Pisa; EU Horizon Europe ITN fellow within the IMPROVE project.
3 - 12.2019	Jacopo Matrone, Master Thesis in Mathematics at the University of Florence, 'Mathematical Model and Numerical Simulation for a Gas Slug Rising in a Volcanic Conduit' .
6.2017 - 6.2019	Federico Brogi, post-doctoral researcher at INGV Pisa.
08.2008 - 12.2011	Davide Giudice, postdoctoral researcher at INGV Pisa.
03.2013 - now	Simone Colucci, postdoctoral researcher at INGV Pisa.
12.2014 - 8.2016	Matteo Bagagli, master student from the Department of Earth Sciences, University of Pisa, at INGV Pisa.

06 - 09.2015	Patrizia Canepa, intern from a computational fluid dynamics course held by DLTM La Spezia at INGV Pisa.

Service

Reviewer for Physics of Plasmas; Physics of Fluids; Journal of Geophysical Research, Geophysical Research Letters, Journal of Volcanology and Geothermal Research, Earth and Planetary Science Letters, Geology, Physics of Fluids, Nature Communications Earth and Environment, Nature Geosciences, Journal of Fluid Mechanics.

Reviewer for NSF - Geophysics panel.

Moderator for EarthArXiv, Earth-Science-focused preprint server.

Editor and member of the Editorial Board for Volcanica.

Member of selection committees for post-doctoral and research scientist positions within and outside INGV; member of MsC thesis committees in Mathematics and Earth Sciences; member of PhD in Statistics thesis jury at University of Geneva, Switzerland.

2021	Organizing Committee for Rittmann Conference for Young Researches 2021, organized by Italian Association for Volcanology (AIV). Editor of the associated abstract volume doi:10.13127/misc/59.
	Member of European Geosciences Union (EGU), American Geophysical Union (AGU), International Association of Volcanology and Chemistry of the Earth's Interior (IAVCEI) and Associazione Italiana di Vulcanologia (AIV).
2018 - 2019	Science Committee member of GMPV (Geochemistry, Mineralogy, Petrology and Volcanology) Division of EGU (European Geosciences Union).
2020 - 2022	Member of the Steering Committee for the IAVCEI Commission on Volcanic and Igneous Plumbing Systems (VIPS) - responsible for Events and Outreach; editor for the eVolcano VIPS contributions.
	Convener of sessions at IAVCEI, AGU and EGU meetings.

Teaching

9 - 12.2006	Tutor for the course Mathematics 0, Faculty of Engineering at the University of Pisa.
3 - 11.2007	Teaching assistant, lab teacher and exams for the course

	General Physics, Degree course in Environmental Sciences and Technologies at the University of Pisa.
10.2008 - 02.2010	Teaching assistant and exams for the course General Physics, Degree course in Information Engineering at the University of Pisa.
2013 - 2014	Guest lecturer for the course "The OpenFOAM framework for the development of numerical models for geophysical applications and research", hosted at the National Institute of Geophysics and Volcanology, Pisa, Italy.
03.2015	NEMOH Short Course "Thermo-fluid Dynamics in Computational Volcanology".
1.2020	Short Course "Fluid Dynamics Modeling in Volcanology" University of Geneva, Doctorate Program for French Swiss Universities (CUSO).
13.5.2025	Seminar *"Application of multiphase flow dynamics to magmatic systems: bubble-melt interactions at different scales"* for the course "Mechanics of Geophysical Fluids" for the Physics curriculum at University of Pisa.
27.3.2026	Short course "Detecting Magmatic Sources from Geophysical Monitoring Signals" University of Geneva, Doctorate Program for French Swiss Universities (CUSO).

Outreach

05.05.2010	School laboratory "Volcanoes: nature's fireworks", for the Museum of Natural History and Territory in Calci, Pisa, Italy.
11 - 18.09.2010	Coordinator of the INGV Volcanological Information Center at Stromboli, Aeolian Islands, Italy.
2014, 2015, 2024	ScienzAperta, INGV-organized outreach activities for schools and public.
2024 - now	Scientific Commission premio Asimov 2023/2024: evaluation of book reviews from students.
2011 - now	European Researcher's Night.
2015 - now	Lessons at primary and secondary schools.
02.2018 - 2026	Memeber of INGV communication working group; responsible for dissemination of volcano science.

Pisa, June 4, 2026